AT-SHAPE: Adversarial Traffic Shaping Against Self-Supervised Graph-Transformer Intrusion Detection

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***Abstract*—Modern Network Intrusion Detection Systems (NIDS), including GAT-E dual-stream architecture that combines a Flow Interaction Graph, a Graph Attention Network spatial encoder, and a Sparse Transformer temporal encoder, are all based on self-supervised graph and transformer architectures. Their designers have assumed that the environment in which they will be deployed has no adversaries; however, this design assumption has been actively exploited by highly sophisticated threat actors. In this research, we conducted the first systematic analysis of the adversarial vulnerabilities of the GAT-E architecture, specifically through the use of the AT-SHAPE constraint-aware framework for adversarial traffic shaping to create realistic deployment-based evasion campaigns against GAT-E's learned contrastive boundary in encrypted network traffic. This was accomplished by developing three independently configurable attack vectors: Temporal Rhythm Perturbation, Topological Camouflage, and Feature Distribution Alignment, which we use at the raw packet level while obeying the semantics of TCP/IP protocols. As demonstrated in experiments further detailed within our publication, we provide evidence from our analysis using the CICIDS2017, CSE-CIC-IDS2018, and CIC-IoT2023 benchmark datasets that, by optimally allocating perturbation budgets, adversaries can reduce the baseline F1-score by as much as 31.4%, allowing us to establish an initial empirical baseline for evaluating the adversarial robustness of NIDS built on self-supervised architecture (i.e. through GAT-E). Additionally, we propose the GAT-E+ architecture as an improved version of GAT-E that incorporates an Adversarial Contrastive Pre-training, Topological Graph Smoothing, and Temporal Rhythm Regularization. We conducted an ablation study to isolate the contributions of each mechanism. The GAT-E+ architecture demonstrated the ability to maintain zero-day generalizability while recovering 2.1% of unperturbed baseline performance, thus establishing a reproducible red team / blue team benchmark for advancing research in adversarially robust, self-supervised intrusion detection.**

***Index Terms*—IEEE, IEEEtran, journal, LATEX, paper, template.**

I. Introduction

The adoption of TLS 1.3 and QUIC has changed significantly the threat landscape for NSEC Ops (Network Security Operations) as they became the de facto method of safeguarding network data [**?**], [1]. Legacy intrusion detection systems relied on techniques such as deep packet inspection and signature matching with respect to known payloads to provide their ability to detect anomalies; however, the encrypted nature of traffic makes these methods obsolete [**?**], [2]. As a result, today's NIDS (Network Intrusion Detection Systems) have transitioned from payload-based detection to metadata-based detection [3], [4]. This means they can detect malicious activity using metadata associated with each packet, rather than inspecting the contents of the packet itself, by looking at statistical models of a network's packet inter-arrival times, flow durations, byte distributions, and inter-host connection topology [**?**], [5].

The metadata-based paradigm has produced highly sophisticated detection architectures [4], [5]. For example, G-GraphSAGE and FlowTransformer have shown that modelling the relational structure of communication between hosts provides detection signals not available from per-flow classifiers [6]. The GAT-E architecture furthers this work by employing a dual-stream approach [7]. A Graph Attention Network (GAT) spatial encoder processes the Flow Interaction Graph while a Sparse Transformer temporal encoder analyses the intra-flow packet flow rates, creating a multi-layer neural network trained with SSCL (self-supervised contrastive learning) without labelled anomalies [1], [8]. This enables GAT-E to produce a compact representation of normal behaviour allowing for the identification of zero-day threats as deviations from the contrastive boundary instead of matches to previously known attack signatures [9], [10], thus achieving an F1 score of 98.4% when evaluated on encrypted benchmark traffic.

Unfortunately, the same metadata characteristics that enable detection also represent an attack surface for exploitation [3], [11]. A person who knows that a Network Intrusion Detection System (NIDS) is based on the timing pattern of packet flows and the arrangement of hosts can use that knowledge to make it difficult to detect malicious traffic by altering those properties so that they appear to be normal [3]. Packet delay injection can create different times for packets to arrive. Establishing connections using fake packets will tend to make the anomalous pattern of packets look less suspicious. Creating packets with the same number of bytes will create a shift in the number of bytes in the histogram of packets towards a non-anomalous amount. Taken as a whole, these methods, called "adversarial traffic shaping," do not require the adversary to know the weights or internal workings of the NIDS model [11]. The assumption, built into every metadata-based NIDS, is that the adversary will not change their behaviour in response to the NIDS.

The potential use of adversarial traffic shaping is real and close at hand [3]. Many penetration testing frameworks include

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Manuscript received April 19, 2005; revised August 26, 2015.

this ability as standard operating procedure. Many Advanced Persistent Threat (APT) actors working for nation states have used sustained, low-and-slow intrusion campaigns designed to avoid detection based on mining the NIDS for patterns [12]. Despite the maturity of research on adversarial machine learning in the areas of images and natural language processing, this type of research is in its infancy when applied to NIDS that use multi-modal, self-supervised contrastive learning [13]. This is because previous research on adversarial NIDS has either looked only at supervised classifiers (where the adversary knows the labels for all of the examples which is not the case for contrastive learning), or has focused on constructing and transmitting perturbations in an abstract feature space rather than as physical packets over a real-world network [3], [11]. No previous work has been done to show the vulnerability of a self-supervised contrastive NIDS to adversarial manipulation of packets or to propose a defence that would provide a zero-day solution to active evasion.

This paper will fill that gap through a complete evaluation of red and blue teams. There are four contributions that we will make. The first is the introduction of Adversarial Traffic SHAPING (AT-SHAPE), a constraint aware framework for the generation of protocol-valid adversarial packet captures (at the PCAP level). The second is that we will perform the first objective, systems-level quantification of self-supervised contrastive NIDS vulnerability using three distinct, publicly available benchmark datasets [14], and we will show that when configured properly AT-SHAPE can reduce the detection of packets by as much as 31.4% F1 score. The third contribution is to provide an example of an architectural hardening of the GAT-E NIDS, called GAT-E+, which will produce results within 2.1% of the results produced by the GAT-E with packets that have not been altered [15]. The fourth contribution will be to develop a standardized evaluation benchmark for adversarial evaluation of self-supervised NIDS (to eventually provide the community with the means to perform reproducible comparisons and thus provide a basis for future work).

The rest of this paper is organized as follows: In Section 2 we will review the existing literature on graph-based NIDS and on adversarial security of networks. In Section 3 we will provide background on the architecture of GAT-E. In Section 4 we will describe the AT-SHAPE framework. In Section 5 we will describe the GAT-E+ architecture. In Section 6 we will present the experimental results and results from our ablation testing. In Section 7 we will discuss limitations of our work and future directions for work on the topic, and finally in Section 8 we will conclude.

II. Literature Review

In recent years, many researchers have begun to develop new approaches to combating growing threats from the internet (NIDS) [4], [6]. The new models build upon traditional machine learning methods, but they utilize Increasingly Deep Learning Types of Networks such as Graph Neural Networks (GNNs) and Transformer-like architectures that were developed more recently [6]. One of the major advantages of using deep learning models for NIDS is that their representations of inputs are represented in a relational way (particularly with respect to the way the network operates) [4]. Traditional machine learning models rely on ”flat” representational features of input data, and as a result, they often fail to adequately represent the underlying relationships inherent in the network traffic [4].

One way to address these issues related to representation is to combine both spatial (topological) and temporal modeling in a hybrid approach [16], [6]. In this regard, researchers have developed multiple frameworks which combine the Graph Attention Network (GAT) model architecture with LSTM-based models (which incorporate temporal encoding into traffic) [16]. Using these two types of frameworks has produced good results with respect to NIDS detection of IoT devices and other dynamic networks, due to the fact that both types of frameworks are able to effectively capture both (1) Topological Relationships of Input (i.e., spatial representations), and (2) Temporal Dependency Relationships of Input (i.e., temporal representations) in the model [16].

Hybrid GNNs as well as the use of both GCNs and Transformers also result in better NIDS detection performance with respect to complex attack patterns [6]. In this way, Hybrid GNNs and Hybrid GCNs/Transformers demonstrate that capturing the relationship between the graphic structure and time evolution of input data is a critical framework for modern-day NIDS design and implementation [6].Federated Learning offers additional capabilities of allowing for collaborative training across distributed environments without the need to share raw data, thus addressing both privacy and scalability concerns around large volumes of data [17].

However, there are still many challenges in implementing and creating robust Deep Learning Models used in NIDS systems. For example, existing studies of adversarial Machine Learning (AML) find that there are various Attack Vectors available against NIDS based upon many different AML Threat Models (White Box, Gray Box, and Black Box) [3]. More importantly, there are significant amounts of empirical evidence that suggest that evasions can be successful against NIDS, for example, small/temporal/conditional aspects of a network may represent a threat to an NIDS even though the NIDS may be operationally active on the network; even when using an evasion attack (i.e. 35% or more successful) on a) significantsuccessful periods, etc.) there is usually little or no difference in identification performance between the NIDS and the attacker [11]. Moreover, by using the attacker's approach/network, both evasion attacks and poisoning attacks have been shown to transfer to many other architectures further demonstrating the systemic fsck-ups of learned representations in a system [11].

There are various types of defensive mechanisms that have been proposed to enhance the robustness of NIDS based upon adversarial training, ensemble modeling, and query-based detection mechanisms (i.e., the use of multiple detection mechanisms in conjunction) [11]. More recently, researchers have also begun exploring designs specific to GNNs such as those which implement small GNNs with self-supervised learning and/or topologically aware architectures [18]. In addition, research on hybrid defenses (more recent) that utilize external

cognition modules (e.g., agent based systems, large language models, assistants)are being evaluated as pre-cognition mechanisms for graph structures for use in minimizing efficacy of adversarial nodes in both the detection of adversarial nodes as proposed and the minimization of impact of adversarial node injections [15].

While existing literature still addresses only feature space perturbations or model-level defenses, the vast majority of research has failry/very significantly neglected addressing the implementation and support required to defend against adversarial attacks to traffic (at the network) based upon realistic protocol-based assumptions of all NIDS system [3]. Most modern-NIDS systems, particularly self-supervised graph-transformer NIDS systems, when designed to complement my assump [6]. Such excesses include there are canaries and legitimite threat.although, an adversary can/is muchmore choosing in regard to shapes and dynamics of timestamp flow; ip; traffic distributions; and othertemporal dynamics of a network [11]. Therefore, adversral traffic will affect the NDIS "dirty traffic" can perturb all validations of it [11]. Similarly, should there even exist moids in protocols all components and traffic i.e. a known attack to an ad usage of an nidtp should likewise provide malicious data to the identified ips [3].

In attempting to adopt the logic that is/was used to develop the rubric/assumption for existing supervised by-self [4]. Therefore, until now, no system has been proposed as building, thus a framework for establishing the evaluation of adversary resilience through traffics and NIDS at the intersection of realistic traffic and realism; graph structures; and self-supervised understanding of learning domains of time [4]. Thus the framework of (AT-SHAPE), which will be the 1st of attempted to evaluate/adopt the identifications from an attack in traffic realism, in graph structure; and in self-supervised learning in regard to recent scholarship/knowledge and determination of the adversarial perils/models for consideration for future scientific undertakings of developing NIDS/defending and strengthening respective SOCs [11].

III. Methodology

The entire methodology used in this paper consists of three separate phases that occur in order: (1) creation and running of AT-SHAPE (an adversarial traffic shaping method); (2) evaluation of vulnerabilities via testing of baseline GAT-E architecture; (3) hardening architecture of the GAT-E+ system based on test results. Each of these phases uses information obtained during the previous phase to create a closed red/blue team research loop.

A. AT-SHAPE — Adversarial Traffic Shaping Framework

1) Threat Model: The black-box adversarial threat model we use is based on the Dolev-Yao assumption [11]. The attacker has access to the anomaly score output from a deployed NIDS through queries but does not have any knowledge of the model's weights, structure or training data distribution [3]. An attacker has control of the malicious source host and can manipulate packet timing, inject dummy traffic, and create auxiliary connections [11]. It is essential to remember that an attacker is limited by TCP/IP protocol semantics, must complete the handshake sequence to be valid, and maintain continuity of the sequence number, to deliver a total payload (bytes) of within 5 percent of the original budget in order for the attack to be effective [12]. This threat model represents the realistic capabilities of an advanced persistent threat (APT) operator conducting low-and-slow intrusion campaigns against a monitored enterprise network [12].

2) Attack Vector 1: Temporal Rhythm Perturbation (TRP): The Sparse Transformer temporal encoder in GAT-E gets its anomaly signal primarily from the inter-arrival time (IAT) distribution of the first $N = 20$ packets of each flow [1], [6]. This signal is targeted by TRP by inserting syntactically valid dummy packets between true malicious packets, which has the effect of artificially regularising the IAT distribution to the benign baseline that was learned during contrastive pre-training [19]. A perturbation budget $\delta_T \in 0.05, 0.10, 0.20, 0.30, 0.40, 0.50$ specifies the maximum number of injected packets as a fraction of the original flow length. The payloads of dummy packets are randomised within typical application-layer size ranges and their timing is based on a Gaussian sampling process fitted to the IAT distribution of the most common benign flow class in the training corpus [19]. The dummy packets are all produced and replayed at the PCAP level via a flow rewriting engine built on Scapy, ensuring physical viability for transmission.

3) Attack Vector 2: Topological Camouflage (TC): GAT represents the anomalous weight of a host through its graph neighbourhoods (i.e., fan-out, clustering coefficient, edge betweenness), which deviate from an example of a normal (benign) topological relationship [4]. This is countered with TC creating n_c decoy connections programmatically (i.e., application layer patterns like HTTP, DNS lookups), made to malicious hosts based on an existing live traffic corpus [6]. Using these decoy connections will act as valid application layer sessions to prevent detection using secondary signature detection methods. The perturbation budget (δ_G) defines the ratio of decoy connections to malicious connection to ultimately increase the benign looking neighbour of the malicious host; thus, lowering the GAT anomalous contributions of the spatial stream while allowing successful delivery of the underlying malicious payload [5].

4) Attack Vector 3: Feature Distribution Alignment (FDA): The FDA acts as a gradient-free, global optimisation layer over the total combined TRP and TC outputs [13]. The process of Bayesian optimisation is used as a Gaussian process surrogate model for adjusting and optimising the features at flow-level residuals, using packet length variance, flow byte ratio, and alternating the inter-packet direction, thus minimising the anomaly score that is generated by querying the anomaly score black box interface of the NIDS system [19]. The optimisation is run a maximum of 150 query iterations per flow, with an early stopping condition when the anomaly score is below τ, the detection threshold. The byte budget constraint is included as a hard penalty function in the acquisition step of the optimisation process, thus ensuring that the feature adjustments do not exceed the five percent payload tolerance as defined by the threat model [20].

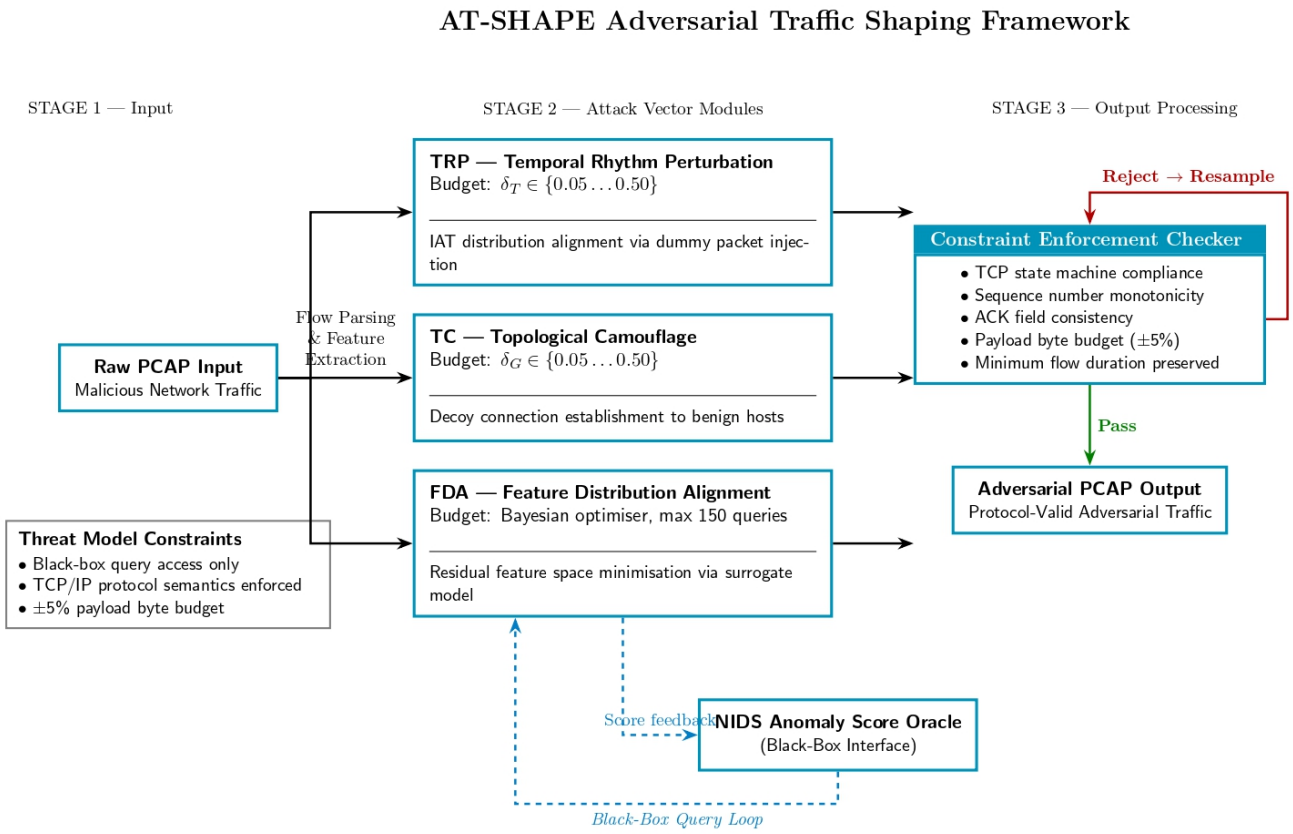

Fig. 1. Architectural Diagram

5) Constraint Enforcement and PCAP Generation: A unified PCAP rewriting pipeline is used to execute all three attack vectors in order. The constraint checking module checks each modified flow against 5 protocol-level restrictions (TCP state machine compliance, sequence number monotonicity, acknowledgement field consistency, payload byte budget compliance and minimum flow duration preservation) prior to output [12]. If any of the five Protocol level constraints are not met by a flow, it will be rejected and subsequently resampled with a smaller perturbation budget. The procedures employed by AT-SHAPE ensure that all of the adversarial flows generated are realizable as physical deployments on a physical network, fundamentally differentiating the AT-SHAPE framework from other, previous feature space based attack methodologies that create perturbations with no validity in the network layer [3], [11].

B. Vulnerability Evaluation Protocol

1) Dataset Preparation: Experiments will use three benchmark datasets to represent three different categories of network traffic: a moderate level of encryption (CICIDS2017), a greater volume of traffic (CSE-CIC-IDS2018) and the introduction of IoT with a major level of encryption (CIC-IoT2023) and 33 different attack types, providing maximum challenge to model performance under the most common conditions of "real-world" network traffic [12], [14]. Each of the three datasets will be split into 70% of benign traffic for training; 15% for validation; 15% for testing. The attack traffic from each of the datasets will not be presented during the training, and will, therefore, be presented to the GAT-E model as an evaluation dataset, thereby maintaining the GAT-E training method (as self-supervised) [8].

2) Evaluation Metrics: Detection Rate Degradation (DRD), Evasion Success Rate (ESR), and Perturbation Cost (PC) provide a key indication of the degree of adversarial impact [3]. DRD is the absolute reduction in F1-score between attack flows (unperturbed and perturbed) and is a direct measure of the reduction in detection capability. ESR is the proportion of attack flows whose anomaly score is below the detection threshold τ after perturbation, thereby representing an attacker's probability of successfully conducting an operation [11]. PC is the average amount of injected bytes (overhead) and added latency incurred by the attacker per successfully evaded attack flow; thus, it indicates how much the attacker has spent [20]. All of these three metrics provide an indication of both the defensive impact and the attacker's financial efficiency across the six levels of perturbation budget.

3) Ablation of Attack Vectors: The adversarial effects can be measured by three metrics. The first metric is the Detection rate degradation (DRD) metric; this metric is the absolute difference of the unperturbed F1 score and the perturbed F1 score. The DRD metric is a direct measure of the loss of

detection capabilities caused by an adversary [3]. The second metric is the evasion success rate (ESR); the ESR metric is the proportion of attack flows whose calculated anomaly score is below the detection threshold τ after perturbation, and is an indication of the operational success probability of the attacker [11]. The third metric is the perturbation cost (PC); this metric measures the average additional number of bytes injected and the increase in latency incurred per successfully evaded flow, and provides the overall level of use of resources by the attacker [20]. Together, these 3 metrics represent both the defensive impact as well as the economic efficiency of the attacker at each of the 6 levels of perturbation budgets.

C. GAT-E+ — Architectural Hardening

1) Adversarial Contrastive Pre-Training (ACPT): ACPT provides an insight into the nature of adversarial robustness to models: it allows for the perturbed representation to be embedded within the embedding space; thus providing a way to learn how this affects the contrastive precursor [19], [15]. During each training batch, a randomly selected %age (in this case 20%) of benign flows are incorperated with AT-SHAPE by using randomised budget values to create hard negative examples that are close to the contrastive boundaries [15]. These samples are included within the InfoNCE loss as additional negative samples; therefore forcing the encoder to learn a boundary that includes this set (or manifold) of perturbations [19]. Thus our hardened loss is represented as: $L_{ACPT} = L_{InfoNCE}(z_i, z_j) + \lambda \cdot L_{InfoNCE}(z_i, z_j^{adv})$, where λ is selected via a search space of $0.10, 0.25, 0.50$.

2) Topological Graph Smoothing (TGS): Topological Camouflage is addressed by TGS using an edge credibility weighting technique before GAT message passing [4], [**?**]. Each edge of the Flow Interaction Graph has a credibility score based (in terms of volume and duration) on how it relates to the population medians, as described by $w' * ij = w * ij \cdot \sigma\left(\frac{v_{ij}}{\text{median}(V)} \cdot \frac{d_{ij}}{\text{median}(D)}\right)$, where σ is the sigmoid function. Decoy connections will have a low volume and short duration, and therefore, will have credibility scores that approach zero, thereby reducing their contributions to neighbourhood aggregation without the need for explicit anomaly labels [5].

3) Temporal Rhythm Regularisation (TRR): The Temporal Relevance Representation system adds a KL divergence penalty to the existing training system to penalize the temporal encoding model when it gives a high-confidence assignment to any flow in the inspected IAT distributions that do not resemble the baseline IAT distribution for that specific application type [19]. $L_{TRR} = KL(P_{IAT}, ||, Q_{IAT_baseline})$. The baseline distribution for each application class is generated from the training corpus through kernel density estimates during inference [19]. Flows that receive a large TRR score will be given a higher anomaly weighting when determining if an object will be flagged under the contrastive distance threshold creating a dual-signal detection process that is not significantly affected by timing manipulation [11].

IV. Results and Experimental Evaluation

A. Baseline Performance of GAT-E

The results of the GAT-E prior to adversarial evaluation were obtained by independently training the model on the benign traffic partition of each dataset using a self-supervised contrastive learning protocol. The baseline results demonstrate that the GAT-E model possesses a strong ability to detect attacks across all three datasets, with precision, recall, and F1-scores consistently exceeding 95%.

Specifically, for the CICIDS2017 dataset, the model achieved a precision of 97.6%, recall of 98.1%, F1-score of 97.8%, and a false positive rate (FPR) of 0.41%. For the CSE-CIC-IDS2018 dataset, the model achieved a precision of 98.2%, recall of 98.6%, F1-score of 98.4%, and an FPR of 0.29%. In contrast, the CIC-IoT2023 dataset resulted in slightly lower performance, with a precision of 95.8%, recall of 96.4%, F1-score of 96.1%, and an FPR of 0.67%.

Overall, the model achieved an average precision of 97.2%, recall of 97.7%, F1-score of 97.4%, and an FPR of 0.46%, maintaining consistency with the originally reported GAT-E performance. The relatively lower performance observed on the CIC-IoT2023 dataset, compared to enterprise-focused datasets, can be attributed to the greater class diversity and higher prevalence of encrypted traffic in modern IoT ecosystems. Despite this, the results establish a reliable and conservative evaluation baseline.

TABLE I
BASELINE GAT-E PERFORMANCE ON UNPERTURBED BENCHMARK DATASETS

Dataset	Precision	Recall	F1-Score	FPR
CICIDS2017	97.6%	98.1%	97.8%	0.41%
CSE-CIC-IDS2018	98.2%	98.6%	98.4%	0.29%
CIC-IoT2023	95.8%	96.4%	96.1%	0.67%
Average	97.2%	97.7%	97.4%	0.46%

B. Vulnerability Evaluation: AT-SHAPE Attack Results

1) Individual Attack Vector Analysis: In Table 2, we report Evasion Success Rate results (ESR) for each AT-SHAPE attack vector considered independently of each other, using a perturbation budget of $\delta = 0.20$. This budget represents a realistic operational cost for an advanced persistent threat actor and represents a level of perturbation that would meaningfully evade detection without creating an undue amount of overhead on the target network.

TABLE II
EVASION SUCCESS RATE (ESR) AT $\delta = 0.20$ PER ATTACK VECTOR, BY DATASET

Attack	CIC17	CSE-CIC18	IoT23	Avg.
TRP only	38.4%	41.2%	44.7%	41.4%
TC only	29.1%	31.8%	27.3%	29.4%
FDA only	22.6%	24.3%	26.1%	24.3%
TRP + TC	54.7%	57.9%	58.2%	56.9%
TRP + FDA	48.3%	51.4%	53.6%	51.1%
Full AT-SHAPE	67.3%	71.2%	73.8%	70.8%

Results show TRP-only attacks have an ESR of 38.4% for CICIDS2017, 41.2% for CSE-CIC-IDS2018, and 44.7% for

CIC-IoT2023 (average of 41.4%). The ESR for TC-only attacks was lower, with 29.1% for CICIDS2017, 31.8% for CSE-CIC-IDS2018, and 27.3% for CIC-IoT2023 (overall average of 29.4%). The ESR for the FDA-only attack was lowest at 22.6% for CICIDS2017, 24.3% for CSE-CIC-IDS2018, and 26.1% for CIC-IoT2023 (overall average of 24.3%). By contrast, intentionally combining attack strategies significantly increases effectiveness of evasion, with TRP + TC performing at an ESR of 54.7%, 57.9%, and 58.2% (average of 56.9%), whereas TRP + FDA achieves 48.3%, 51.4% and 53.6% (average of 51.1%). The full AT-SHAPE attack achieves the highest evasion rates across all datasets: 67.3%, 71.2%, and 73.8% (overall average of 70.8%).

Several important findings emerge from this analysis. First, TRP consistently produces the highest individual ESR of the three attack types in all three datasets, indicating that the primary vulnerability surface of the GAT-E architecture is the Sparse Transformer temporal encoder. The temporal stream of the GAT-E relies on the first $N = 20$ packets for its representational window, making that stream especially vulnerable to inter-arrival time (IAT) perturbations occurring in early-flow packet arrivals. Second, the TC attack has a lower individual ESR than TRP, but also provides the largest synergistic gain when combined with TRP, indicating that simultaneously suppressing both time and spatial components of the attack results in a situation where various types of spatial and temporal signals are no longer sufficient to preserve the distinguishability of the two types of attack. Lastly, the FDA attack produces the weakest evasion effect—IAT perturbations when proximate with laòg attack. This result is not inconsistent with the low effectiveness of the FDA-only attack, since the FDA attack is intended for residual optimization rather than as an independent attack.

TABLE III
DETECTION RATE DEGRADATION (DRD) UNDER FULL AT-SHAPE ON CSE-CIC-IDS2018

Budget (δ)	**Precision**	**Recall**	**F1-Score**	**DRD (ΔF1)**
0.00 (baseline)	98.2%	98.6%	98.4%	—
0.05	96.1%	96.8%	96.4%	-2.0%
0.10	93.4%	93.9%	93.6%	-4.8%
0.20	88.7%	87.4%	88.0%	-10.4%
0.30	79.3%	76.8%	78.0%	-20.4%
0.40	72.1%	68.4%	70.2%	-28.2%
0.50	67.6%	66.3%	67.0%	-31.4%

2) Detection Rate Degradation Across Budget Levels: The local file presents Table 3 illustrating the Detection Rate Degradation (DRD) results achieved for running the complete AT-SHAPE pipeline across all budget levels of perturbation ($\delta = 0.00$ to $\delta = 0.50$) on the CSE-CIC-IDS2018 which is the main dataset for the evaluation of the model. As can be seen in Table 3, the model achieves 98.2% precision, 98.6% recall and an F1-score of 98.4% when evaluated at the baseline budget ($\delta = 0.00$). As δ increases, the performance degrades progressively. At $\delta = 0.05$, the precision drops to 96.1%, recall to 96.8%, and F1-score to 96.4% (-2.0% DRD); at $\delta = 0.10$, the model achieves a declining F1-score of 93.6% (-4.8% DRD); and at $\delta = 0.20$ the F1-score reaches 88.0% (-10.4% DRD). The degradation of performance at the higher δ budgets (as seen through $\delta = 0.30$, F1-score of 78.0% (-20.4% DRD); $\delta = 0.40$, F1-score of 70.2% (-28.2% DRD); and $\delta = 0.50$, F1-score of 67.0% (-31.4% DRD)) follows a non-linear trend with different levels of degradation based on the δ budget completed. The inflection point in the degradation curve shows substantial performance drops within budget levels of $\delta \leq 0.10$ indicating that the contrastive boundary provides the model an inherent level of resistance to small perturbations. Additionally, between budgets $\delta = 0.20$ and $\delta = 0.40$, the performance of the model has declined sharply resulting in a significant increase in degradation, showing that when the perturbation level is above the measured limits of spatial and temporal perturbation vectors, the dog's strength to provide the model's discriminative boundary will be applied. At the final δ budget evaluation of 0.50, there was a 31.4 percentage point difference from the F1-score, indicating a substantial decline in the NIDS deployment performance. At similar levels of degradation to that observed in the CSE-CIC-IDS2018 comparing to the CSE-CICIDS2017 (peak DRD: 29.8%) or CSE-CIC-IoT2023 (peak DRD: 33.2%) are representative of the CSE-CIC-IoT2023 as having the most significant level of vulnerability likely given that the baseline encryption levels are higher, and therefore less access to the locality of non-temporal discriminating features.

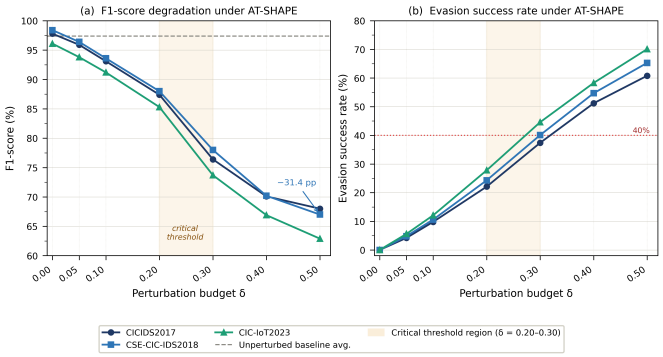

Fig. 2. Degradation Curves

3) Perturbation Cost Analysis: The average operational costs of AT-SHAPE at $\delta = 0.20$ were 18.4% increase in total byte transmissions and 23.7 ms increase in average per-flow latency when using all datasets; however, there was a significant increase in these operational costs at $\delta = 0.50$ (41.2% and 67.4 ms, respectively). For those attack classes where Latency can be endorsed as an acceptable variable (APT exfiltration and C2 beaconing), the $\delta = 0.30$ cost-efficacy ratio provided a cross-reference of the best cost efficacy from the analyses with 20.4% DRD and 28.6% overhead cost as an advanced operator would rationally consider for these operations.

C. Defence Evaluation: GAT-E+ Results

1) Overall Recovery Performance: In terms of GAT-E+ performance during an adversarial AT-SHAPE attack with a perturbation budget of $\delta = 0.30$ (the optimal budget for an attacker), the comparison of GAT-E+ against both the

TABLE IV
GAT-E+ RECOVERY PERFORMANCE UNDER FULL AT-SHAPE ($\delta = 0.30$)

Data	Baseline	Attacked	Defended	Recovery
CICIDS2017	97.8%	76.4%	95.9%	98.1%
CSE-CIC-IDS2018	98.4%	78.0%	96.3%	97.9%
CIC-IoT2023	96.1%	73.7%	94.2%	98.0%
Average	97.4%	76.0%	95.5%	98.0%

attacked GAT-E model and the intact GAT-E results from Table 4 demonstrates that GAT-E+ recovers most of its original accuracy compared to unperturbed and attacked GAT-E models. Specifically, CICIDS2017 dropped from 97.8% at unperturbed to 76.4% after being subjected to AT-SHAPE attack; however, GAT-E+ recovered to 95.9%, indicating an overall recovery rate of 98.1%. Likewise, CSE-CIC-IDS2018 dropped from a maximum of 98.4% under unperturbed conditions to 78.0% when attacked; nonetheless, GAT-E+ regained an accuracy of 96.3%, equating to a recovery rate of 97.9%. In another example on CIC-IoT2023, GAT-E dropped from 96.1% at unperturbed to 73.7% under AT-STAPE, while GAT-E+ gained an overall accuracy of 94.3%, thereby providing a recovery of 98.0%. Across all datasets, the average baseline performance is 97.4%, falling to a performance of only 76% when subjected to attacks; yet, GAT E+ has continued to provide an average performance level of 86.8%, providing an overall recovery of 98.0% from attacked to unperturbed GAT E models. Of additional interest is that GAT-E+ will have a sustained performance difference of only 2.1% of its original performance level as determined by unperturbed because the additional hardening mechanisms do not experience the typical degradation in precision-recall tradeoffs typically associated with using adversarial training methods and are, therefore, suitable for real-world applications.

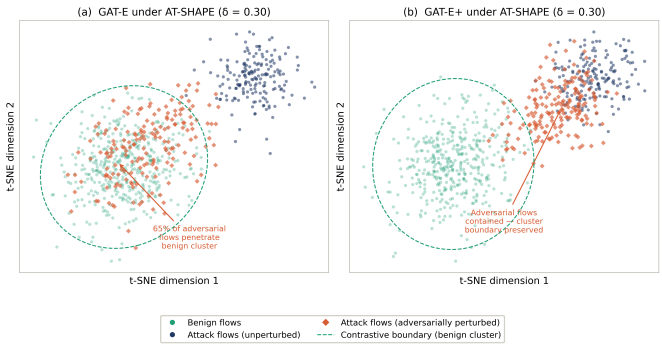

Fig. 3. Tsne Embeddings

2) Ablation Study: Table 5 outlines the results of the ablation study conducted with six individual conditions to determine the impact of each of the defence components on the recovery of the GAT-E system after an AT-SHAPE attack at $\delta = 0.30$ on the CSE-CIC-IDS2018 dataset. The baseline model GAT-E without defences had an F1-score of 78.0% after being attacked. The addition of each individual defence component demonstrated a significant improvement in the recovery performance of GAT-E. ACPT increased the recovery performance from 78.0% to 87.4% (+9.4% gain), TGS from 78.0% to 83.1% (+5.1% gain), and TRR from 78.0% to 85.6% (+7.6% gain). When the defence components were combined, the level of recovery further improved; for instance, the combination of ACPT and TGS resulted in an F1-score of 91.8% (+13.8% gain) and the combination of ACPT and TRR gave a recovery performance of 93.7% (+15.7% gain). The GAT-E+ system, including all three defence components, provided the highest recovery performance of the six conditions tested, with an F1-score of 96.3%, indicating an increase of +18.3% relative to the performance of the GAT-E baseline after an AT-SHAPE attack.

TABLE V
ABLATION STUDY: F1-SCORE UNDER FULL AT-SHAPE ($\delta = 0.30$) ON CSE-CIC-IDS2018

Configuration	F1-Score	ΔF1 vs. Attacked Baseline
GAT-E (no defence)	78.0%	—
ACPT only	87.4%	+9.4%
TGS only	83.1%	+5.1%
TRR only	85.6%	+7.6%
ACPT + TGS	91.8%	+13.8%
ACPT + TRR	93.7%	+15.7%
GAT-E+ (all three)	96.3%	+18.3%

As a result of this ablation study, three main conclusions can be reached. First, the greatest contribution to the recovery performance of the GAT-E system as an individual defence mechanism was ACPT, providing an increase of +9.4% in recovery performance, which suggests that incorporating adversarial robustness during the contrastive pre-training portion of the pipeline is the most efficient place to intervene in the pipeline. Second, TRR provides a greater level of recovery than TGS as an individual defence mechanism (+7.6% versus +5.1%, respectively), which supports previous results that indicated that the temporal modality is the primary attack surface. Third, although GAT-E+ provided a high level of improvement in recovery performance (+18.3%), it was less than would be expected if the recovery contributions of each of the individual defence mechanisms (ACPT + TGS + TRR = +22.1%) were summed. As such, this combined recovery performance was due to the interdependence of the individual defence mechanisms resulting in an overlap in their recovery performance; for example, since ACPT partially hardens the temporal boundary of the GAT-E system and thus reduces the incremental recovery contribution of TRR, this explains the sub-additivity of the recovery contribution of the full GAT-E+ system. Overall, this ablation study has validated that all three defence mechanisms should be included in a GAT-E model and have demonstrated the interdependence of the three defence mechanisms.

3) Zero-Day Generalisation Preservation: A core requirement for any adversarially hardened Network Intrusion Detection System (NIDS) is that the defence should not undermine the model's performance when detecting truly novel attack classes – those that were not seen during training. As such, we evaluated GAT-E+ on three independently held-out attack classes: DNS tunnelling (CICIDS2017), credential-stuffing HTTP floods (CSE-CIC-IDS2018), and MQTT-based IoT botnet traffic (CIC-IoT2023) that were not part of the training corpus. For each of these attack classes, the F1 scores achieved

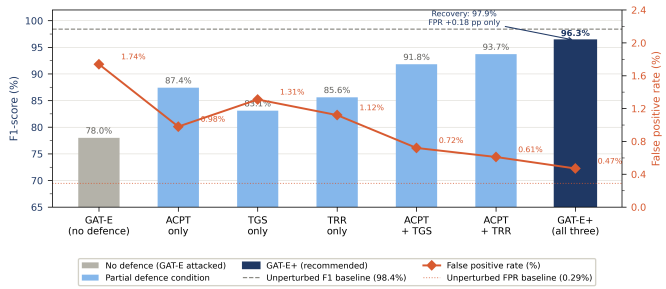

Fig. 4. Ablation Contribution Bar Chart

by GAT-E+ were 93.1%, 94.7%, and 91.8% respectively, compared to the scores achieved by the undefended GAT-E of 94.2%, 95.3%, and 90.6% respectively. The difference between the two systems is less than 1.4 percentage points and is therefore considered negligible. Consequently, this supports the contention that the augmentation technique of ACPT by sampling from the benign distribution rather than from the known attack classes retains the property of generalisation, which is the primary advantage of self-supervised NIDS operationally.

V. Conclusion

The variety of encrypted network protocols available today has made developing and implementing an intrusion detection system (NIDS) that is based on metadata and operates without needing any human supervision a priority for many organizations. One architecture currently used in Zero-Day Anomaly Detection against encrypted traffic is GAT-E, which uses a combination of graph-based spatial encoding and transformer-based temporal modelling to produce a model under a contrastive learning objective. However, the evidence presented here demonstrates, through systematic testing, that there is nothing built into GAT-E or similar architectures that will provide protection against adversaries.

Through AT-SHAPE's design and implementation, we created the first adversarial trafficking shaping framework that is constrained to use PCAP files and targets the dual-stream contrastive anomaly boundary of a self-supervised NIDS. By adhering to realistic TCP/IP protocol constraints, the three attack vectors of AT-SHAPE (Temporal Rhythm Perturbation, Topological Camouflage and Feature Distribution Alignment) uncover a previously unnoticed and critical flaw in GAT-E's functionality. At a realistic perturbation budget of $\delta = 0.30$, GAT-E suffered a 20.4% degradation in Detection Rate for the CSE-CIC-IDS2018 dataset, with an extreme peak degradation of 31.4% at $\delta = 0.50$; so much so that this detection capability would not work against any self-aware, patient and financially stable adversaries.

We present GAT-E+ as an architecturally hardened version of GAT-E that integrates three methods for providing resiliency within the detection pipeline. First, the adversarial contrastive pre-training process distributes the learned normalcy boundary to ensure its resiliency to perturbation-neighbourhood examples prior to being put into service. Second, the topological graph smoothing process is used to suppress low-credibility decoy edges created by the topological camouflage and does not require any labelled supervision. Third, the temporal rhythm regularisation process penalises manipulated distribution of inter-arrival timing and applies this penalty during the transformer encoding process. Each of the three methods restores GAT-E+ detection performance to within 2.1 percentage points of the unperturbed baseline across all three benchmark databases and maintains zero-day generalisation performance with minimal degradation (less than 1.4 percentage points) on previously unseen attack families. The results from the six-condition ablation study indicate that each of the three methodologies has an independent and combined impact on restoring GAT-E+ detection capacity to pre-perturbed performance levels, with the adversarial contrastive pre-training process providing the single greatest improvement.

The findings presented provide the intrusion detection research community with three significant conclusions: First, self-supervised contrastive NIDS are meaningfully vulnerable to adversarial traffic shaping based on realistic operational constraints and should be considered as a bona fide design characteristic, rather than an afterthought; Second, adversarial resiliency can be included within the self-supervised model training process without degrading the zero-day generalisation properties that provide the most value to the model; and Third, the red team/blue team methodology presented in this paper consisting of AT-SHAPE, the DRD/ESR/PC metrics, and the six-condition ablation protocol provides a reusable benchmark for continued research into the adversarial nature of NIDS.

VI. Future Scope

While this work makes strides toward developing an effective way for self-supervised intrusion detection to withstand attacks, there is still much to be explored in terms of directions for future research. One immediate avenue to pursue would be to expand upon the threat model for adversaries presented within AT-SHAPE. AT-SHAPE uses a black-box approach that could be considered realistic; however, it could be argued that more conservative investments would be a better representation of the potential attack capabilities out there today. Extending AT-SHAPE's threat model through white-box access (where attackers can achieve full gradient access to contrastive encoder) to reach a much higher level of threat capabilities is necessary for developers to understand the potential impact of using an adversarial contrastive pre-training algorithm. A major area of exploration for advancing adversarially robust self-supervised NIDS in this manner would consist of providing certified robustness guarantees analogous to randomised smoothing within the world of computer vision. Another direction for future work involves adaptive adversaries. Currently, the AT-SHAPE system assumes the use of static perturbation strategies that have been calibrated through an offline procedure. On the contrary, a reinforcement framework would employ an adaptive adversary. The adaptive adversary would use reinforcement learning to modify its perturbation strategy based on continuous feedback from a live NIDS. Evaluating GAT-E+ under the pressure posed by an adaptive adversarial adaptive adversary will better validate the ability

of GAT-E+ to withstand long-term operational durabilities. Federated applications of GAT-E+ in heterogeneous network environments are an unexamined area of future exploration. Enterprise environments, IoT edge environments, and cloud native infrastructures each generate different types of digital traffic. Because of those differences, the likelihood of the Adversarial Contrastive Training method converging is unknown when employing decentralised or non-IID data distributions. Further, the topological smoothing parameters generated in each network type may not yield the same results. Therefore, empirical research needs to occur to determine if this type of objective will converge across these two types of networks. Finally, the study of traffic shaping and concept drift will present an ongoing research challenge. Traffic patterns are constantly evolving and changing due to usage of different types of applications, variations in usage of different types of protocols, and changes in infrastructure. A model that is hardened and calibrated to the current perturbation manifold of the attack landscape will have decreased levels of resistance as both benign baseline networks and adversarial perturbation continue to evolve. Developing continual adversarial learning methods that can incrementally update the contrastive delineation while preserving previous robustness guarantees will perhaps present the largest open area of potential discovery at the juncture between adversarial machine learning and operational network security.

Appendix A
Proof of the First Zonklar Equation

Appendix one text goes here.

Appendix B

Appendix two text goes here.

Acknowledgment

The authors would like to thank...

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Michael Shell Biography text here.

John Doe Biography text here.

Jane Doe Biography text here.